AIKODEX

Stadium Creator 2

Procedural grandstands and GPU-animated crowds for Unity



User Manual · Version 2.0 · Unity 6.6+ · Universal Render Pipeline

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1. Welcome

Thanks for picking up Stadium Creator 2. The idea behind it is simple: building a convincing crowd should take minutes, not weeks. You lay out grandstands with a few sliders, and the package fills them with thousands of individually varied people who sit, stand, clap, cheer and react to your game.

The crowd is made from 49 realistic adult characters in a wide range of ages, body types and outfits. Spectators are recoloured into your team kits so a stand can be a sea of home shirts with the odd away fan, and every person plays their own animation at their own pace, so no two neighbours move in lockstep.

All of this runs on the GPU. There is no GameObject, Animator or material per spectator, which is why a stadium of 20,000 people costs your CPU almost nothing.

2. Requirements and installation

- **Unity 6.6** or newer.
- The **Universal Render Pipeline** (URP 17.6 or newer), Forward or Forward+.
- A graphics API with **compute shader** support. Stadium Creator 2 was developed and tested on Windows with Direct3D 11. WebGL is not supported because it has no compute shaders.

Import the package from the Package Manager. Everything lives in the `Stadium Creator 2` folder; you can move that folder anywhere in your project, as the tools find their data automatically.

Folder	Contents
Data	The character library (meshes and textures for all 49 characters) and the animation database. These two files are most of the package size.

Demo	A small demo scene with one curved bleacher.
Documentation	This manual and the Getting Started guide.
Editor	Menus and inspectors.
Materials	Concrete, wood, painted steel, aluminium and seat materials for the grandstands, plus the crowd material.
Runtime	The components you add to your scenes.
Shaders	The crowd compute shader and the crowd surface shader.

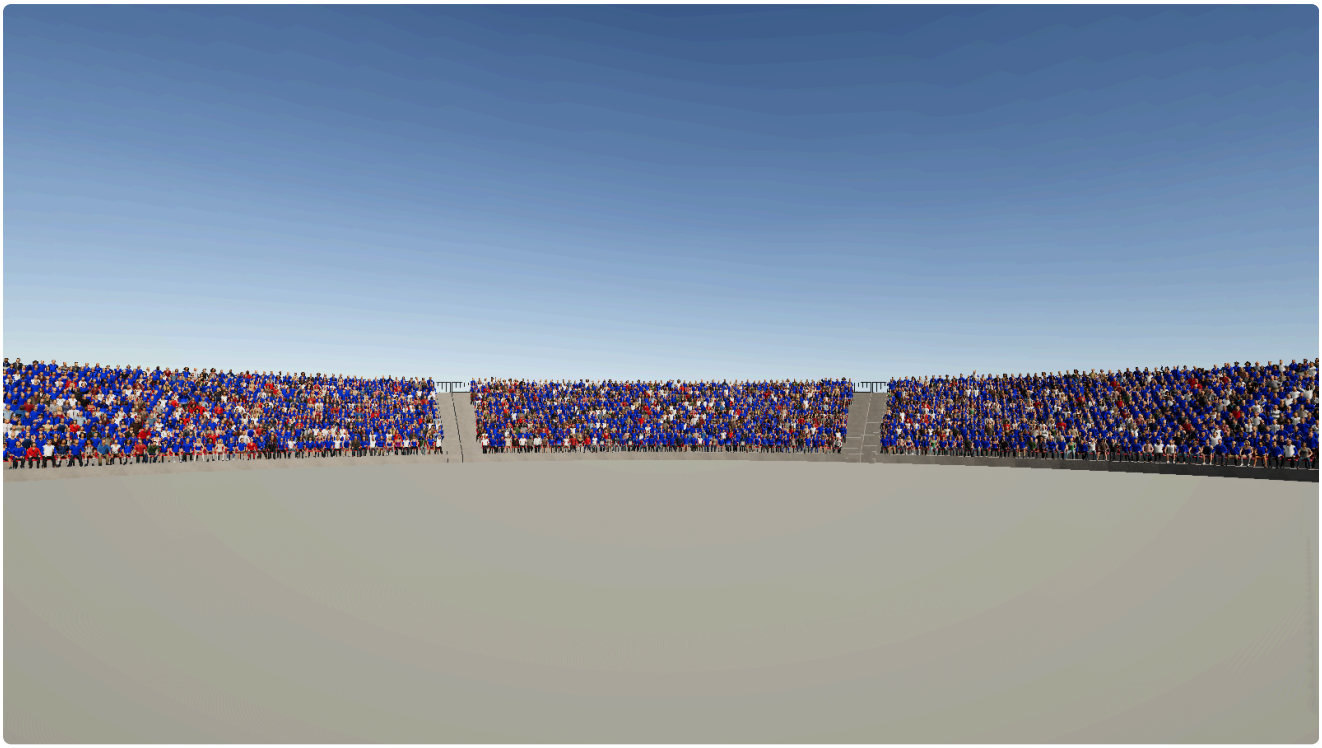
3. Your first stadium

Choose **Tools** ▸ **Stadium Creator 2** ▸ **Create Stadium**, or right-click in the Hierarchy and pick **Stadium Creator 2** ▸ **Stadium**. This creates two things: a **Stadium** object, which owns the crowd, and a **Grandstand** child with 5 rows of 10 seats. The new stadium is placed on the ground in front of the Scene view camera.



What Create Stadium gives you: one Classic grandstand of 5 × 10 seats, already populated.

From here you mostly work with grandstands. Select one and change its settings; it rebuilds instantly and the crowd re-seats itself. To grow the stadium, click **Add Grandstand Beside** on a grandstand (or **Add Grandstand** on the Stadium). The new stand copies the one before it and continues along the same front edge and curve. Keep clicking and the stands wrap into a bowl. Where two stands meet, the side railing in between is dropped automatically and the aisle stairs form a natural walkway.



Curved Classic grandstands with concrete tiers, seen from the field.

Stands are ordinary GameObjects. Move, rotate or delete them freely; the crowd follows. Different styles can be mixed in the same stadium, for example a concrete main stand opposite temporary steel bleachers.

4. Grandstand reference

Each grandstand is a single procedural mesh plus a list of seats. Spectators face the local −Z direction, towards the field, and rows climb towards +Z.

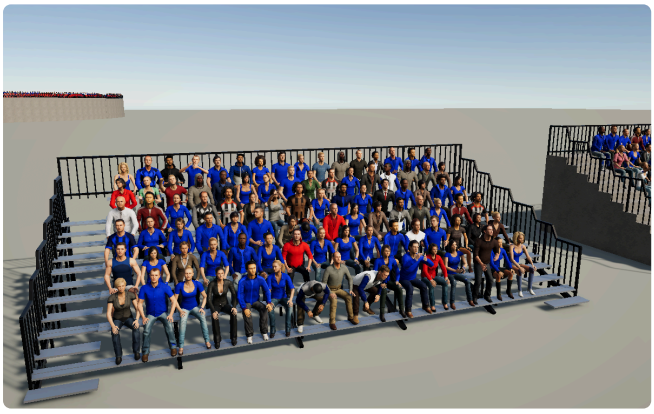
Styles



Classic: stepped tiers (concrete or wood) with plastic bucket seats.



Wooden Bleacher: plank benches, foot boards and a slatted riser on a steel frame.



Steel Bleacher: aluminium planks on an open steel frame.



Concrete Slabs: people sit on the step edge with their feet on the step below.

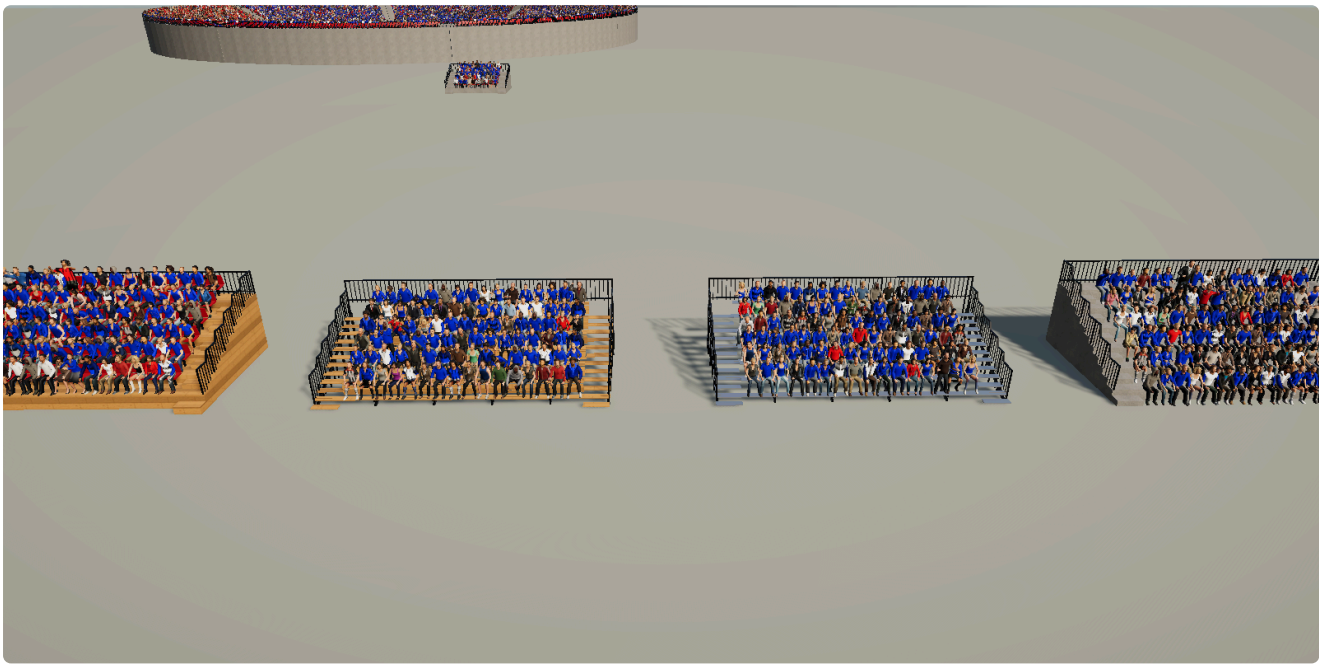
Settings

Setting	What it does
Rows	Number of rows, 1 to 60.
Columns	Seats in the front row, 2 to 200. When a stand curves around the field the rows behind are longer and gain extra seats, just as in a real bowl.
Curve	Bends the stand. Positive values wrap it around the field, negative values bend it away. At ±1 the front edge is a half circle. Very strong negative curves are limited so the back rows never fold over themselves.
Style	One of the four styles above.
Deck	Classic only: concrete or wooden tiers.

Railings	Railing along the back and down both open sides. Posts and bars are spaced to fit the stand exactly.
Aisle Stairs	Adds a stairway at both ends of every row. The seating area is narrowed to make room.
Seat Colour	Classic only: the colour of the bucket seats.
Seat Spacing	Distance between neighbouring seats, 0.45 to 1.5 m. Increase it for a more relaxed, less crowded look.
Auto Row Size	Uses the recommended row depth and rise for the style. Untick it to set Row Depth and Row Rise yourself. For Concrete Slabs the rise always equals the seat height, because the step is the seat.
Density	Fraction of seats that are occupied in this stand. Combined with the Stadium's density.
Team	−1 lets the Stadium decide, 0 fills the stand with home fans, 1 with away fans.
Standing Bias	Extra chance that people in this stand stand up. Good for a lively home end.
Materials	The concrete, wood, steel, aluminium and seat materials. Replace them with your own to match your art style. Textures are mapped in metres, so any tiling texture works.

5. Stadium reference

The Stadium object (the `CrowdManager` component) owns the crowd. It collects seats from every grandstand below it, decides who sits where, and renders everyone. The inspector lists the stadium's grandstands at the top, followed by the crowd and mood settings. Everything else sits under *Advanced*.



All four styles side by side, each populated by the same Stadium.

Setting	What it does
Density	Fraction of all seats that are occupied.
Home Ratio	Share of stands given to home fans when a stand's Team is set to automatic. Fans are grouped by stand, with a few away supporters sprinkled in, the way real crowds sit.
Home / Away Team	Primary and secondary kit colours, kit pattern (solid, vertical stripes, hoops, sleeve bands, chest band, halves), how many fans wear the shirt, and an alternate kit worn by some of them.
Seed	Changes who sits where and what they wear, without changing the layout.
Mood buttons	Switch the crowd's mood instantly. Intensity blends between Calm and the chosen mood.
Goal!	An eight-second celebration that spreads out from the stadium's centre, then returns to the previous mood.

Advanced settings

Setting	What it does
Max Spectators	A hard cap on the number of people, useful on low-end hardware. 0 means no limit.

Casual / bottom / hair palettes	Colours used for fans who do not wear the team shirt.
Casual Recolor Ratio	How many non-kit clothes are recoloured from the palettes, for extra variety.
Height Jitter	Small random scale variation between people.
Emotion Profiles	One entry per mood: how many people stand, playback speed, how often people change what they are doing, how quickly they react, and which kinds of animation they prefer.
Default Wave Speed	When above 0, every mood change spreads out from the stadium at this speed in metres per second.
Culling Camera	The camera used for culling and level of detail. Empty uses the main camera (or the Scene view camera in edit mode).
LOD Distances	The distances at which characters switch to simpler meshes, plus the maximum draw distance. They are scaled automatically with the camera's field of view.
Shadow Casting Max LOD	Characters up to this LOD cast shadows. The default of 0 limits shadows to the nearest people.
Render In Edit Mode	Shows the crowd in the Scene view without entering Play mode.

6. Moods and scripting

Everything lives in the `AiKodex.StadiumCreator2` namespace. The crowd understands eleven moods: *Calm, Interested, Anticipating, Happy, Excited, Cheering, Ecstatic, Disappointed, Sad, Angry* and *Celebrating*. A mood is not a single animation. It changes how likely people are to stand, which animations they choose and how energetically they play them. When the mood changes, each person reacts after a short, personal delay, so the change rolls through the stands instead of switching like a light.

```
using AiKodex.StadiumCreator2;
using UnityEngine;

public class MatchEvents : MonoBehaviour
{
    public CrowdManager homeStand;
    public Transform goalMouth;

    public void OnAttackBuildsUp()  => CrowdManager.SetEmotion(CrowdEmotion.Anticipating, 1f);
    public void OnNearMiss()        => CrowdManager.TriggerReactionAll(CrowdEmotion.Disappointed,
1f, 5f);

    public void OnGoal()
    {
        // celebrate for 8 seconds, spreading from the goal at 40 m/s, then return to the
previous mood
        homeStand.TriggerReaction(CrowdEmotion.Celebrating, 1f, 8f, goalMouth.position, 40f);
    }
}
```

Method	Description
CrowdManager.SetEmotion(mood, intensity)	Static. Sets the mood of every stadium in the scene. Intensity 0 to 1 blends from Calm.
CrowdManager.SetEmotion(mood, intensity, origin, speed)	Static. As above, but the reaction spreads from a world position at the given speed.
CrowdManager.TriggerReactionAll(mood, intensity, seconds)	Static. A temporary reaction for every stadium, which then returns to the previous mood.
stadium.ApplyEmotion(mood, intensity)	The same as SetEmotion for a single stadium.
stadium.TriggerReaction(mood, intensity, seconds, origin, speed)	A temporary reaction for one stadium, optionally spreading from a point.
stadium.Generate()	Rebuilds the crowd immediately, for example after changing grandstands from code.

stadium.RequestRegenerate()	Rebuilds the crowd a moment later. Several changes in a row are batched into one rebuild.
grandstand.Rebuild()	Rebuilds a grandstand after you change its fields from code (rows, columns, curve, style…).

Grandstands can be created at runtime too: add a `Grandstand` component under a Stadium, set its fields and materials, and call `Rebuild()`. `PlaceAfter(other)` lines a stand up with the end of another one, exactly like the Add Grandstand button.

7. Custom seating layouts

The crowd does not care where seats come from. Any component that implements `ICrowdSeatSource` and sits under a Stadium is picked up automatically, which lets you put people anywhere: around a boxing ring, on a hillside, along a race track or on a concert floor.

```
public class HillsideSeats : MonoBehaviour, ICrowdSeatSource
{
    public void CollectSeats(List<CrowdSeat> seats)
    {
        // position = where the spectator's feet are, forward = the direction they face
        seats.Add(new CrowdSeat { position = transform.position, forward = transform.forward,
team = -1 });
    }
}
```

`CrowdLineupSource` in the Runtime folder is a complete, minimal example that lines people up on a grid. Seated animations expect a seat surface 0.45 m above the feet. Standing spectators simply stand on the given position.

8. Performance

The CPU cost of a stadium is constant: about 0.05 ms per frame whether it holds 200 or 50,000 people. The GPU cost depends on how many spectators are visible and how close they are. Measured on a laptop RTX 3070 Ti at 2560 × 1440 in the editor:

Scene	Visible spectators	Frame rate
Broadcast view of a full stadium	15,000	about 160 fps
Pitch-side close-up	5,400	about 140 fps
Aerial view, everyone visible	40,000	about 94 fps
Aerial view, everyone visible	54,000	about 70 fps

If you need more headroom:

- Pull the **LOD distances** closer. Distant people use a 450-triangle mesh that is indistinguishable at range.
- Set **Shadow Casting Max LOD** to −1 if the crowd does not need to cast shadows (the stands themselves still do).
- Lower **Density** or set **Max Spectators** on lower-end platforms.
- Prefer a few large grandstands over many tiny ones. The number of stands does not affect crowd cost, but each stand is a separate mesh.

Memory: the character textures take about 160 MB of GPU memory and are shared by every stadium. The animation data takes 6 MB. Each stadium adds roughly 2 KB per spectator in GPU buffers.

9. How it works

Each of the 49 characters was prepared ahead of time with four levels of detail (about 7,400, 3,400, 1,300 and 450 triangles) and a texture atlas. All atlases are packed into one texture array, so the whole crowd shares a single material. The animation clips are stored as compact bone rotations rather than baked vertex positions, so every body type keeps its own proportions while sharing the same motion.

Every frame, a compute shader decides what each person is doing, culls everyone outside the camera view, picks a level of detail and builds the skinning data for visible people only. The crowd is then drawn with two indirect draw calls per level of detail. The CPU only uploads a handful of numbers, which is why the cost does not grow with the crowd.

10. Troubleshooting

The crowd does not appear

Check that the Stadium's inspector does not show a warning about missing character data, and that the grandstands are children of the Stadium. In the Scene view, make sure *Render In Edit Mode* is enabled. On platforms without compute shader support (WebGL), the crowd is disabled and a warning is logged.

The crowd disappears when I look around in the Scene view

If a *Culling Camera* is assigned, culling follows that camera rather than the Scene view. Clear the field to use the active camera.

The grandstand changed but the crowd did not follow

The crowd rebuilds a fraction of a second after a change. If you modify grandstands from code, call `Rebuild()` on the grandstand; the stadium is notified automatically.

People float above or sink into custom seats

Seated people expect a seat surface 0.45 m above the seat position you provide. Lower or raise your seat positions accordingly.

Everyone wears the same colour

Lower *Kit Ratio* on the team kits for more casual clothing, raise *Casual Recolor Ratio* for more variety, or change the *Seed*.

11. Credits and licences

The spectator characters are derived from the **Microsoft Rocketbox** avatar library, released under the MIT licence. The full licence text is in `Third-Party Notices.txt`. Everything else in Stadium Creator 2 was made by AiKodex.

If you have a question or an idea, please get in touch through our publisher page on the Unity Asset Store. Reviews help us a lot, so if Stadium Creator 2 saved you time, we would love to hear about it.